Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. The learning rate was set to 0.01 with the Adam optimizer.

2 Results

Figure 1 shows the validation accuracy during training.

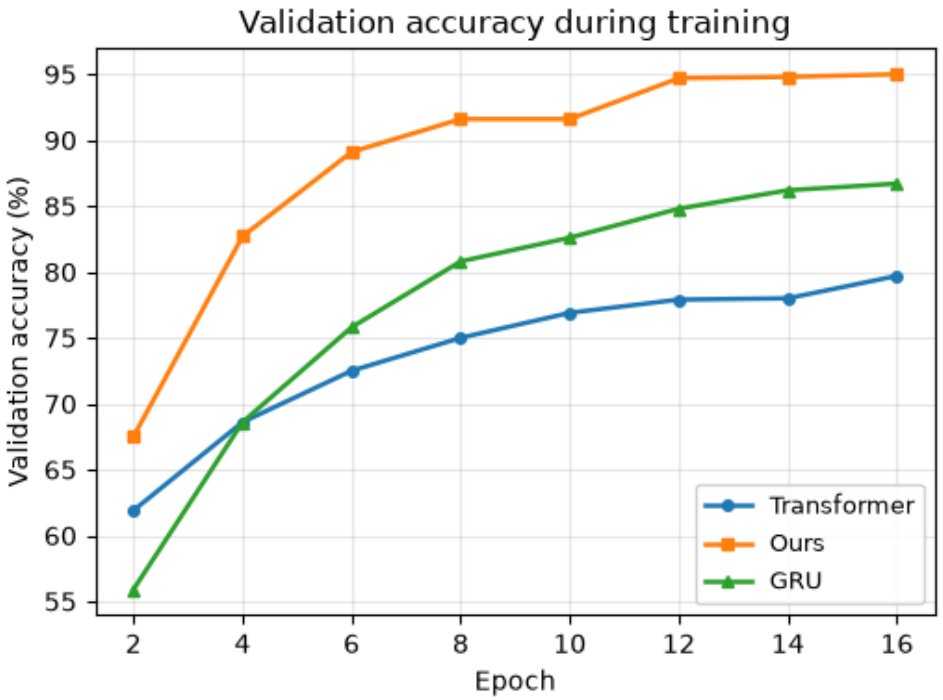


Figure 1: Validation accuracy during training for the models in this experiment.

As shown in Figure 1, Ours outperforms GRU by 13.3 percentage points at epoch 6. GRU reaches 69% validation accuracy at epoch 4. GRU reaches 61.9% validation accuracy at epoch 2.

Figure 2 shows the validation accuracy of Transformer by dataset.

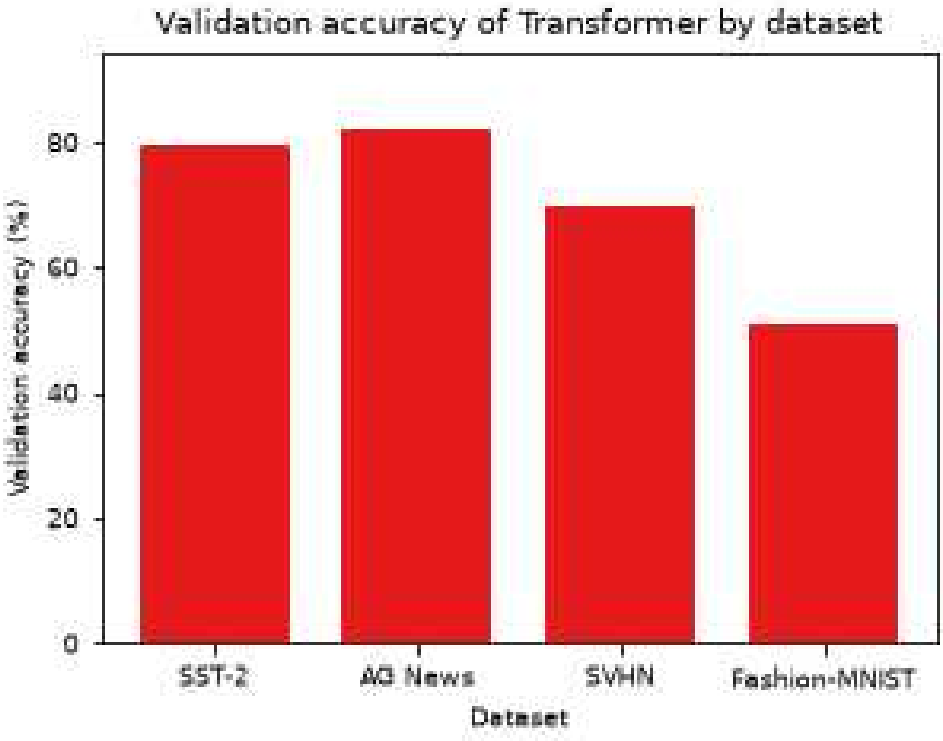


Figure 2: Validation accuracy of Transformer by dataset for the models in this experiment.

Transformer reaches 34.9% validation accuracy on AG News. As shown in Figure 2, Transformer reaches 70.1% validation accuracy on SST-2.

Figure 3 shows the accuracy by dataset.

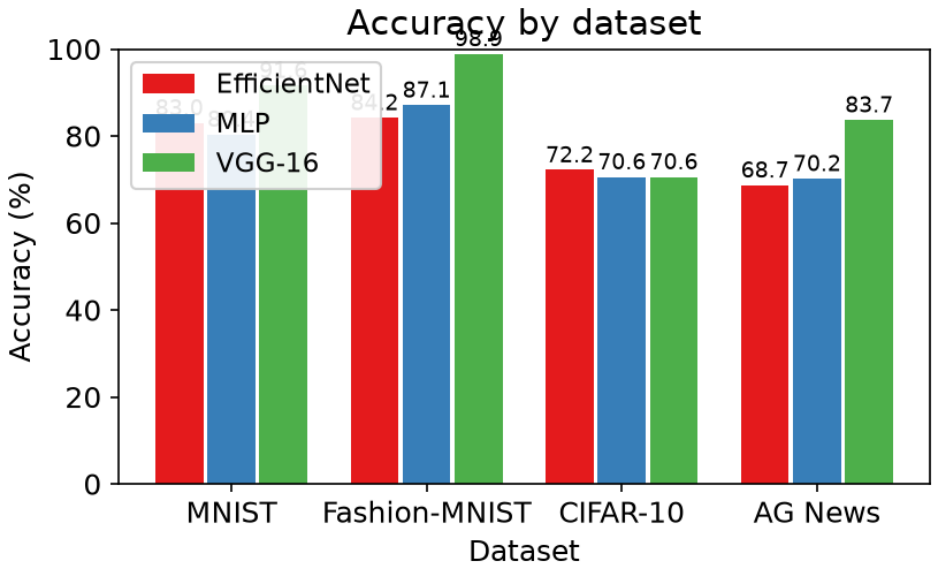


Figure 3: Accuracy by dataset for the models in this experiment.

VGG-16 outperforms EfficientNet by 13.6 percentage points on MNIST (Figure 3). VGG-16 achieves 33.7% accuracy on AG News. EfficientNet reaches 79.2% accuracy on Fashion-MNIST.

Figure 4 shows the accuracy of ViT-S by dataset.

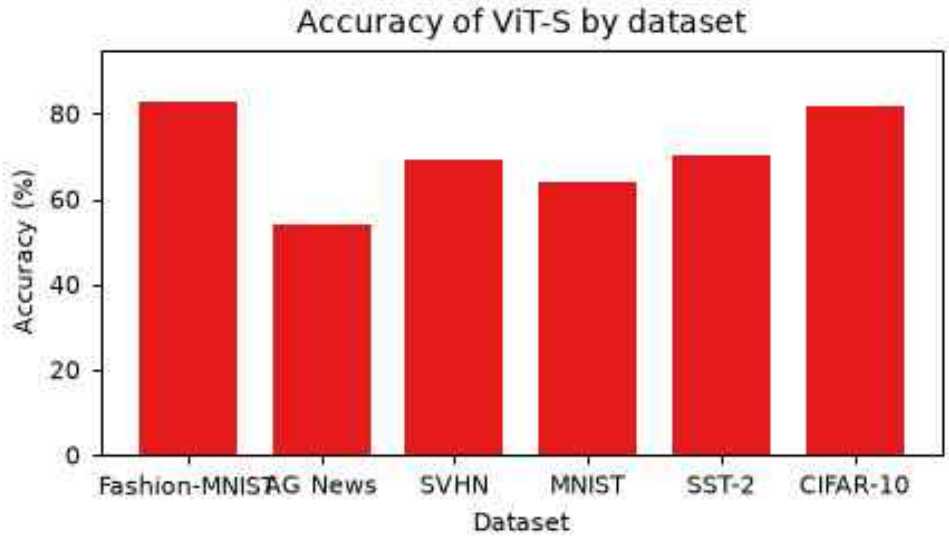


Figure 4: Accuracy of ViT-S by dataset for the models in this experiment.

As shown in Figure 4, ViT-S reaches 91.0% accuracy on CIFAR-10. On CIFAR-10, ViT-S obtains a accuracy of 62.5%. ViT-S scores 20.6 points higher on Fashion-MNIST than on MNIST. ViT-S achieves 65.4% accuracy on SST-2 (Figure 4).

3 Discussion

On Fashion-MNIST, XGBoost obtains an accuracy of 64.5%. Overall, the results are in line with what we expected from the literature.